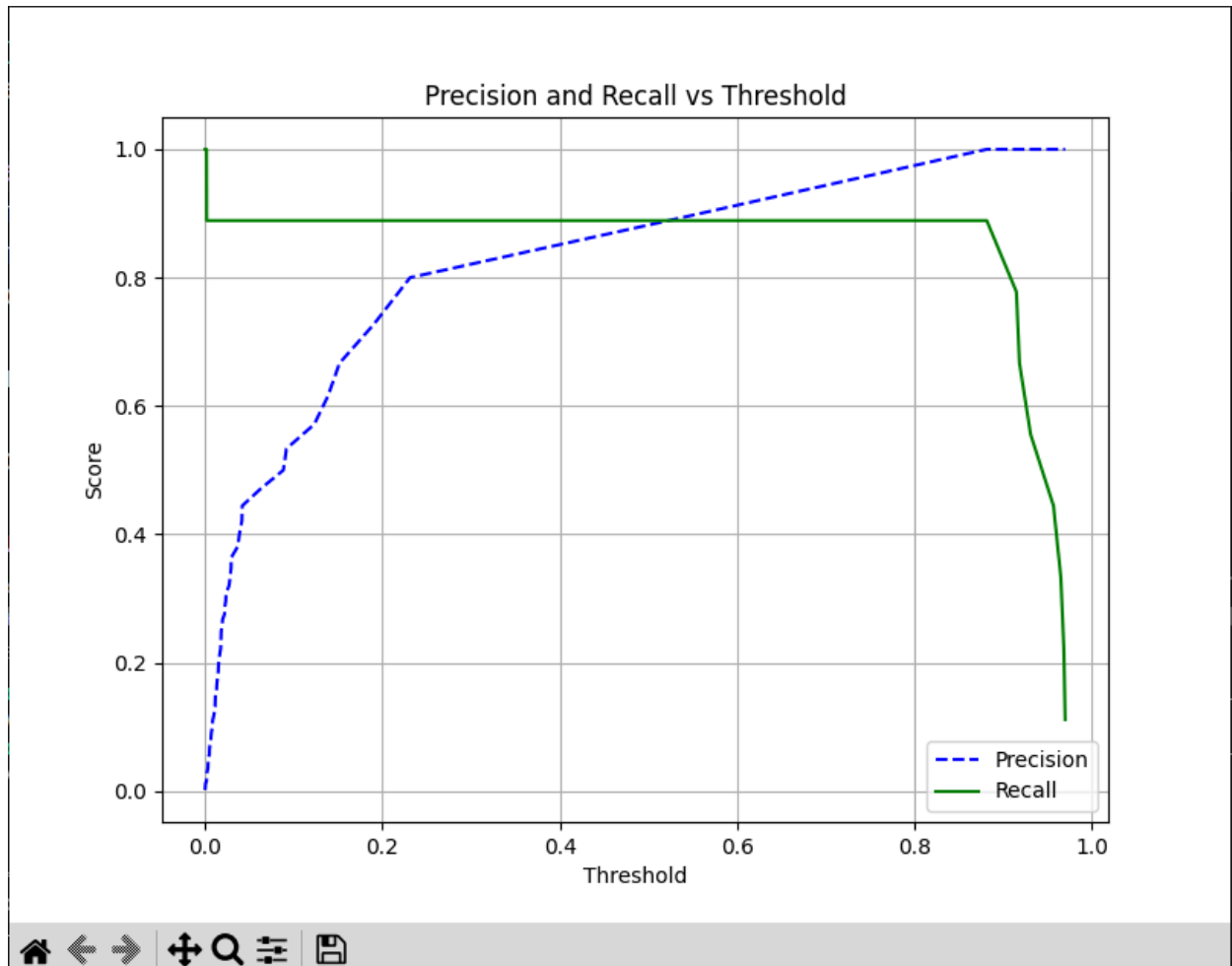


Fraud Detection (classification probleme)

Engineered an end-to-end classification system for fraud detection where the dataset features were anonymized, except for 'Time' and 'Amount'. To prepare the data, I applied a RobustScaler to these specific features to effectively handle outliers. The dataset was highly imbalanced, with legitimate transactions (Class 0) being 570 times more frequent than fraudulent ones (Class 1).

To address this class imbalance, I utilized three models: Random Forest, XGBoost, and CatBoost. I optimized their parameters using GridSearchCV, and then implemented a Voting Classifier among them to ensure robustness and higher accuracy, rather than relying on a single model.

Furthermore, I did not settle for the default classification threshold of 0.5, as the optimal threshold varies based on specific business requirements. Instead, I plotted a Precision-Recall curve to determine the most suitable value, which resulted in an optimal threshold of 0.55.



When evaluated on the test dataset, the model achieved excellent results. The confusion matrix showed that it successfully identified 82 fraudulent transactions out of 97 (with only 15 False Negatives), and accurately classified 56,842 legitimate transactions (with only 21 False Positives). According to the classification report, this resulted in a Precision of 0.80, a Recall of 0.85, and an F1-score of 0.82 for the fraud class, alongside an overall

accuracy of 1.00.

```
confusion matrixs :  
[[56842  21]  
 [  15  82]]  
*****  
classification_report :  
              precision    recall  f1-score   support  
  
      0           1.00       1.00       1.00     56863  
      1           0.80       0.85       0.82         97  
  
   accuracy           1.00  
  macro avg           0.90       0.92       0.91     56960  
 weighted avg           1.00       1.00       1.00     56960
```

From a business perspective, achieving an 85% recall rate significantly prevents financial losses by capturing the majority of fraudulent activities. Simultaneously, the exceptionally low false-positive rate guarantees that genuine customer transactions remain largely undisrupted.